# Evaluation runs: 2pc-ds (two-pc-disel)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.2 | 2 | 1 h, 250 GB | 16 | 2384 | brain03, 2026-11-02 |
| 3 | 104976 | 82944 | conforms | 212.6 | 4 | 1 h, 250 GB | 64 | 2574748 | brain03, 2026-11-02 |

### stateless: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.2 | 2 | 1 h, 250 GB | 16 | 2370 | brain03, 2026-11-02 |
| 3 | 104976 | 82944 | conforms | 208.7 | 2 | 1 h, 250 GB | 64 | 2559952 | brain03, 2026-11-02 |
| 4 | 95883264 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 1.3 | 2 | 1 h, 250 GB | 64 | 25600 | brain03, 2026-11-02 |
| 3 | 104976 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 1.3 | 4 | 1 h, 250 GB | 64 | 25600 | brain03, 2026-11-02 |
| 3 | 104976 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 4 bag correct.t16.p.r3, 4 bag skip-ack-loop.t16.p.r3, 5 bag skip-ack-loop, 5 bag correct.

left out by the table rules (timeout): 4 correct bag.

skipped: a smaller size had reached its limit: 4 skip-ack-loop bag, 4 skip-ack-loop fifo, 5 correct bag, 5 correct fifo, 5 skip-ack-loop fifo, 5 skip-ack-loop bag, 6 skip-ack-loop fifo, 6 correct bag, 6 skip-ack-loop bag, 6 correct fifo, 7 correct fifo, 7 correct bag, 7 skip-ack-loop fifo, 7 skip-ack-loop bag, 8 skip-ack-loop bag, 8 correct fifo, 8 correct bag, 8 skip-ack-loop fifo, 9 skip-ack-loop bag, 9 correct bag, 9 correct fifo, 9 skip-ack-loop fifo.

left out by the table rules (invalid: foreign load): 4 bag correct, 4 bag skip-ack-loop.

# stateful

## stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 104976 | 82944 | conforms | 6.8 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 95883264 | 84934656 | conforms | 2010.3 | 222 | 48 h, 250 GB | brain04, 2026-11-05 |
| 6 | | | (48 h) | | | 48 h, 250 GB | brain04, 2026-11-03 |

## stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 104976 | 82944 | conforms | 7.7 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 95883264 | 84934656 | conforms | 2209.8 | 240 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

## stateful: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 104976 | 2304 | 2304 | violation | 3.9 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 95883264 | 147456 | 147456 | violation | 977.0 | 220 | 48 h, 250 GB | brain04, 2026-11-05 |

## stateful: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 104976 | 2304 | 2304 | violation | 3.8 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 95883264 | 147456 | 147456 | violation | 1115.1 | 236 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

**Not run (stateful).** left out by the table rules (invalid: foreign load): 5 skip-ack-loop bag, 5 correct bag, 6 bag skip-ack-loop, 7 bag correct.

planned but never scheduled (grid points beyond the ladder): 5 skip-ack-loop bag, 6 skip-ack-loop bag, 6 correct bag, 7 skip-ack-loop bag, 7 correct bag, 8 correct bag, 8 skip-ack-loop bag, 9 correct bag, 9 skip-ack-loop bag, 10 correct bag, 10 skip-ack-loop bag, 11 skip-ack-loop bag, 11 correct bag.

left out by the table rules (ok): 4 skip-ack-loop bag, 4 correct bag.

left out by the table rules (timeout): 5 bag skip-ack-loop, 5 correct bag, 5 bag correct, 7 bag skip-ack-loop.

skipped: a smaller size had reached its limit: 6 skip-ack-loop bag, 6 correct bag, 6 correct fifo, 6 skip-ack-loop fifo, 7 skip-ack-loop bag, 7 skip-ack-loop fifo, 7 correct fifo, 7 correct bag, 8 skip-ack-loop fifo, 8 correct bag, 8 skip-ack-loop bag, 8 correct fifo, 9 correct bag, 9 correct fifo, 9 skip-ack-loop bag, 9 skip-ack-loop fifo.

# offline

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 4.4 | 8 | 1 h, 250 GB | 4.4 | 3.7 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 95883264 | 84934656 | conforms | 1329.7 | 412 | 48 h, 250 GB | 1329.7 | 1186.8 | 0.0 | merge | brain04, 2026-11-05 |
| 4 | 95883264 | 84934656 | conforms | 1326.7 | 444 | 1 h, 250 GB | 1326.6 | 1202.7 | 0.0 | merge | brain05, 2026-11-03 |

### offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 0.2 | 8 | 1 h, 250 GB | 0.1 | 0.2 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 4.7 | 8 | 1 h, 250 GB | 4.7 | 3.8 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 95883264 | 84934656 | conforms | 1412.9 | 442 | 1 h, 250 GB | 1412.8 | 1192.9 | 0.1 | merge | brain05, 2026-11-03 |

### offline: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 4.4 | 8 | 1 h, 250 GB | 4.4 | 0.2 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 95883264 | 147456 | 256 | violation | 1329.7 | 226 | 48 h, 250 GB | 1329.7 | 6.7 | 0.0 | merge | brain04, 2026-11-05 |
| 4 | 95883264 | 147456 | 256 | violation | 1326.7 | 244 | 1 h, 250 GB | 1326.6 | 6.4 | 0.0 | merge | brain05, 2026-11-03 |

### offline: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 4.7 | 8 | 1 h, 250 GB | 4.7 | 0.2 | 0.0 | merge | brain05, 2026-11-03 |
| 4 | 95883264 | 147456 | 256 | violation | 1412.9 | 238 | 1 h, 250 GB | 1412.8 | 6.5 | 0.0 | merge | brain05, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 5 bag skip-ack-loop, 5 bag correct, 8 skip-ack-loop bag.

skipped: a smaller size had reached its limit: 5 correct fifo, 5 correct bag, 6 skip-ack-loop bag, 6 correct bag, 6 correct fifo, 6 skip-ack-loop fifo, 7 skip-ack-loop bag, 7 skip-ack-loop fifo, 7 correct fifo, 7 correct bag, 8 skip-ack-loop fifo, 8 correct bag, 8 skip-ack-loop bag, 8 correct fifo, 9 correct bag, 9 correct fifo, 9 skip-ack-loop bag, 9 skip-ack-loop fifo.

refused (an input missing at run time): 5 skip-ack-loop bag, 5 skip-ack-loop fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 5.1 | 175 | 10 min, 250 GB | 0.1 | 5.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 108.2 | 5261 | 48 h, 80 GB | 6.8 | 108.2 | 0.0 | merge | pinky08, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 5.2 | 182 | 10 min, 250 GB | 0.1 | 5.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 107.6 | 5123 | 48 h, 80 GB | 6.5 | 107.5 | 0.0 | merge | pinky07, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 7.5 | 155 | 10 min, 250 GB | 0.1 | 7.5 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 10.5 | 2347 | 48 h, 80 GB | 6.8 | 10.5 | 0.0 | merge | pinky08, 2026-11-07 |

### cross offline, Java implementation against the Rust specification: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 7.8 | 167 | 10 min, 250 GB | 0.1 | 7.7 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 11.4 | 2362 | 48 h, 80 GB | 6.5 | 11.4 | 0.0 | merge | pinky07, 2026-11-07 |

**Not run (cross offline, Java implementation against the Rust specification).** skipped: a smaller size had reached its limit: 5 bag skip-ack-loop, 5 bag correct.

planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 5.3 | 171 | 10 min, 250 GB | 5.1 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 155.7 | 10901 | 10 min, 250 GB | 155.6 | 3.7 | 0.1 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 5.3 | 181 | 10 min, 250 GB | 5.3 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 82944 | conforms | 266.7 | 12242 | 10 min, 250 GB | 266.6 | 3.8 | 0.1 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 5.8 | 171 | 10 min, 250 GB | 5.1 | 0.1 | 0.7 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 155.7 | 10901 | 10 min, 250 GB | 155.6 | 0.2 | 0.1 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 16 | violation | 5.3 | 181 | 10 min, 250 GB | 5.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 104976 | 2304 | 64 | violation | 266.7 | 12242 | 10 min, 250 GB | 266.6 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop, 5 bag skip-ack-loop, 5 bag correct.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 4.2 | 229 | 1 h, 250 GB | | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3 | 104976 | 82944 | conforms | 211.9 | 14692 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 256 | conforms | 4.4 | 200 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 104976 | 82944 | conforms | 218.9 | 15291 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 4.2 | 229 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 104976 | 2304 | 2304 | violation | 207.0 | 14692 | 1 h, 250 GB | 3 | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 400 | 64 | 64 | violation | 4.4 | 200 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 104976 | 2304 | 2304 | violation | 214.0 | 15291 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop, 5 bag skip-ack-loop, 5 bag correct.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 5.1 | 175 | 10 min, 250 GB | 5.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 139.1 | 9257 | 10 min, 250 GB | 139.0 | 3.7 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 5.1 | 182 | 10 min, 250 GB | 5.1 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 265.9 | 10023 | 10 min, 250 GB | 265.9 | 3.8 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.7 | 155 | 10 min, 250 GB | 7.5 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 18.8 | 2191 | 10 min, 250 GB | 18.7 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.7 | 167 | 10 min, 250 GB | 7.7 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 10.0 | 576 | 10 min, 250 GB | 10.0 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 147456 | 147456 | | conforms* | 202.5 | 9300 | 48 h, 200 GB | 202.4 | 8.6 | 0.0 | merge | volta13, 2026-11-07 |

### twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 5.1 | 175 | 10 min, 250 GB | 5.1 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 139.2 | 9257 | 10 min, 250 GB | 139.0 | 0.2 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 5.2 | 182 | 10 min, 250 GB | 5.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 266.2 | 10023 | 10 min, 250 GB | 265.9 | 0.2 | 0.3 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 5.1 | 175 | 10 min, 250 GB | 0.1 | 5.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 139.2 | 9257 | 10 min, 250 GB | 3.7 | 139.0 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 5.1 | 182 | 10 min, 250 GB | 0.2 | 5.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 266.5 | 10023 | 10 min, 250 GB | 3.8 | 265.9 | 0.6 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.5 | 155 | 10 min, 250 GB | 0.1 | 7.5 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 19.2 | 2191 | 10 min, 250 GB | 0.2 | 18.7 | 0.5 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.7 | 167 | 10 min, 250 GB | 0.1 | 7.7 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 10.2 | 576 | 10 min, 250 GB | 0.2 | 10.0 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 147456 | 147456 | | conforms* | 202.5 | 9300 | 48 h, 200 GB | 8.6 | 202.4 | 0.0 | merge | volta13, 2026-11-07 |
| 4 | 84934656 | 147456 | 256 | violation | 1685.9 | 9511 | 48 h, 200 GB | 1685.9 | 202.4 | 0.0 | merge | volta13, 2026-11-07 |

### twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 7.5 | 155 | 10 min, 250 GB | 0.1 | 7.5 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 18.9 | 2191 | 10 min, 250 GB | 3.7 | 18.7 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 7.7 | 167 | 10 min, 250 GB | 0.2 | 7.7 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 10.1 | 576 | 10 min, 250 GB | 3.8 | 10.0 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 7.1 | 162 | 10 min, 250 GB | 7.0 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 142.6 | 9072 | 10 min, 250 GB | 142.5 | 3.7 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 6.2 | 180 | 10 min, 250 GB | 6.2 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 82944 | 82944 | conforms | 161.1 | 9823 | 10 min, 250 GB | 160.8 | 3.8 | 0.3 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 4.3 | 160 | 10 min, 250 GB | 4.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 11.7 | 2948 | 10 min, 250 GB | 11.7 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.2 | 160 | 10 min, 250 GB | 7.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 12.2 | 2893 | 10 min, 250 GB | 12.1 | 0.2 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 147456 | 147456 | | conforms* | 210.2 | 9587 | 48 h, 200 GB | 210.2 | 8.6 | 0.0 | merge | volta13, 2026-11-07 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 7.2 | 163 | 10 min, 250 GB | 7.0 | 0.1 | 0.3 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 142.6 | 9072 | 10 min, 250 GB | 142.5 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 6.2 | 180 | 10 min, 250 GB | 6.2 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 160.8 | 9823 | 10 min, 250 GB | 160.8 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 7.0 | 162 | 10 min, 250 GB | 0.1 | 7.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 142.5 | 9072 | 10 min, 250 GB | 3.7 | 142.5 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 6.2 | 180 | 10 min, 250 GB | 0.2 | 6.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 160.8 | 9823 | 10 min, 250 GB | 3.8 | 160.8 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 4.4 | 160 | 10 min, 250 GB | 0.1 | 4.3 | 0.2 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 11.7 | 2948 | 10 min, 250 GB | 0.2 | 11.7 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 7.1 | 160 | 10 min, 250 GB | 0.1 | 7.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 12.2 | 2893 | 10 min, 250 GB | 0.2 | 12.1 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 4.4 | 160 | 10 min, 250 GB | 0.1 | 4.3 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 11.7 | 2948 | 10 min, 250 GB | 3.7 | 11.7 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 16 | violation | 7.1 | 160 | 10 min, 250 GB | 0.2 | 7.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 64 | violation | 12.2 | 2893 | 10 min, 250 GB | 3.8 | 12.1 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 4 fifo skip-ack-loop.

planned but never scheduled (grid points beyond the ladder): 4 bag correct.

refused (an input missing at run time): 4 bag skip-ack-loop.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 8.7 | 174 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 192.4 | 10872 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 8.4 | 168 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 82944 | 82944 | conforms | 211.1 | 11897 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 8.6 | 166 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 16.7 | 2602 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 8.1 | 171 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 2304 | 2304 | conforms | 17.6 | 2482 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 64 | violation | 7.5 | 178 | 1 h, 250 GB | 192 | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 2304 | violation | 21.4 | 2491 | 1 h, 250 GB | 312 | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **skip-ack-loop**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 64 | violation | 7.7 | 165 | 1 h, 250 GB | 168 | brain01, 2026-11-03 |
| 3 | 82944 | 2304 | 2304 | violation | 21.8 | 2445 | 1 h, 250 GB | 199 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 4.0 | 205 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 82944 | 82944 | conforms | 177.5 | 12510 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 256 | conforms | 4.0 | 196 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 82944 | 82944 | conforms | 207.9 | 11112 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model bag, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 3.4 | 158 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 2304 | 2304 | conforms | 10.2 | 762 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 64 | 64 | conforms | 3.4 | 154 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 2304 | 2304 | conforms | 10.8 | 3697 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model bag, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 64 | violation | 4.0 | 205 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 82944 | 2304 | 2304 | violation | 172.6 | 12510 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair skip-ack-loop, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 256 | 64 | 64 | violation | 4.0 | 196 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 82944 | 2304 | 2304 | violation | 203.1 | 11112 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** planned but never scheduled (grid points beyond the ladder): 4 bag correct, 4 bag skip-ack-loop.